# DSC630Week3.2

September 27, 2025

[1]:
```python
import matplotlib
import matplotlib.pyplot as plt

import pandas as pd
import numpy as np

import seaborn as sns
```

Improving a Dodgers Marketing Promotion - Exploratory Data Analysis in Python

Objective: Determine what night would be the best to run a marketing promotion to increase attendance.

Data source:

Dodgers Major League Baseball data from 2012

Importing and Previewing Data

[4]:
```python
# Loading data into a Pandas DataFrame
dodgers_data = pd.read_csv("dodgers-2022.csv")
```

[7]:
```python
# Checking dimensions
print(dodgers_data.shape)
```

```
(81, 12)
```

[9]:
```python
# Previewing data
dodgers_data.head(5)
```

[9]:
```
  month  day  attend day_of_week opponent  temp   skies day_night cap shirt  \
0   APR   10   56000     Tuesday  Pirates    67  Clear          Day  NO    NO
1   APR   11   29729   Wednesday  Pirates    58  Cloudy       Night  NO    NO
2   APR   12   28328    Thursday  Pirates    57  Cloudy       Night  NO    NO
3   APR   13   31601      Friday   Padres    54  Cloudy       Night  NO    NO
4   APR   14   46549    Saturday   Padres    57  Cloudy       Night  NO    NO

  fireworks bobblehead
0        NO         NO
1        NO         NO
2        NO         NO
```

```
3       YES         NO
4        NO         NO
```

[11]:

```python
# Displaying summary information for string columns

print("String Data:\n")
print(dodgers_data.describe(include=['O']))
```

```
String Data:

       month day_of_week opponent   skies day_night cap shirt fireworks  \
count     81          81       81      81        81  81    81        81
unique     7           7       17       2         2   2     2         2
top      MAY     Tuesday  Giants  Clear      Night  NO    NO        NO
freq      18          13        9      62        66  79    78        67

       bobblehead
count          81
unique          2
top            NO
freq           70
```

[13]:

```python
# Displaying summary information for numeric columns
dodgers_data.describe()
```

[13]:

| | day | attend | temp |
|---|---|---|---|
| count | 81.000000 | 81.000000 | 81.000000 |
| mean | 16.135802 | 41040.074074 | 73.148148 |
| std | 9.605666 | 8297.539460 | 8.317318 |
| min | 1.000000 | 24312.000000 | 54.000000 |
| 25% | 8.000000 | 34493.000000 | 67.000000 |
| 50% | 15.000000 | 40284.000000 | 73.000000 |
| 75% | 25.000000 | 46588.000000 | 79.000000 |
| max | 31.000000 | 56000.000000 | 95.000000 |

[15]:

```python
# Checking missing data sums
dodgers_data.isna().sum()
```

[15]:

```
month          0
day            0
attend         0
day_of_week    0
opponent       0
temp           0
skies          0
day_night      0
cap            0
shirt          0
```

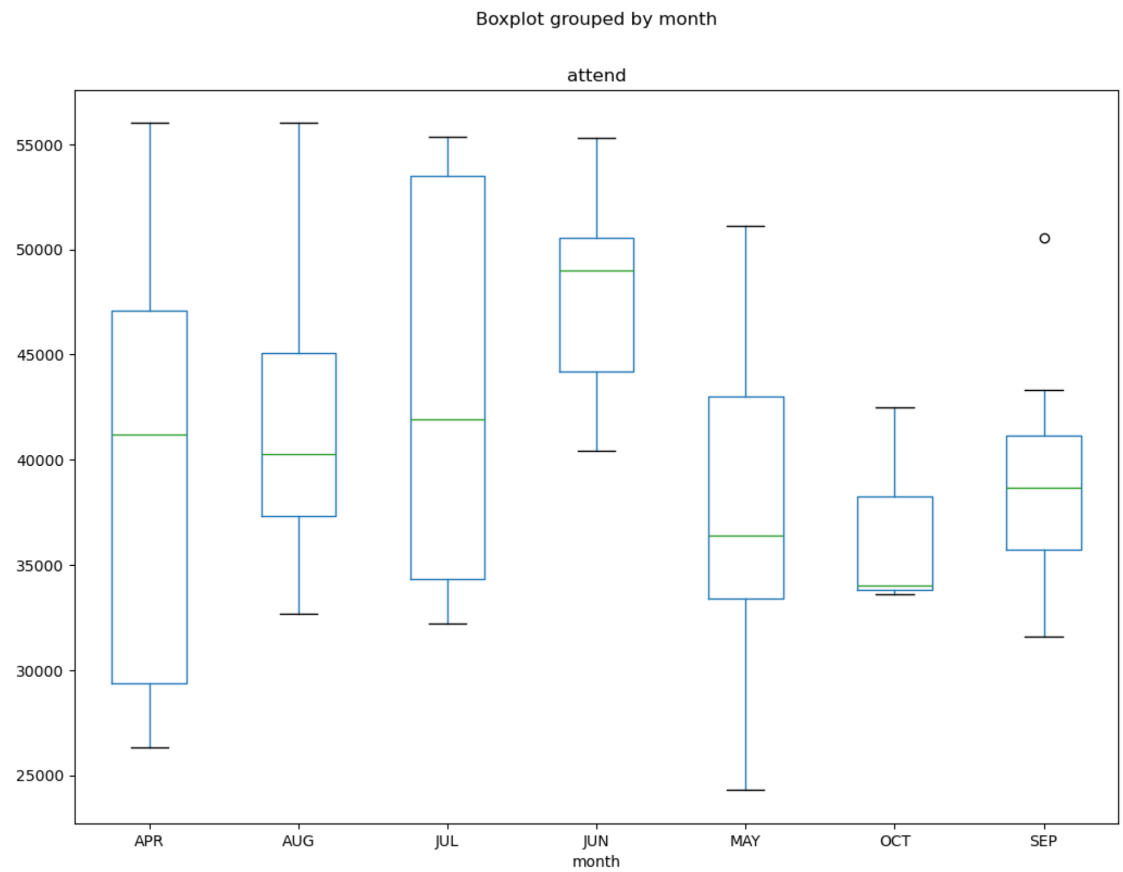
```
fireworks      0
bobblehead     0
dtype: int64
```

Exploratory Visualizations

[18]:
```
# Boxplot of monthly attendance

dodgers_data.boxplot(by ='month', column =['attend'], figsize=(12,9), grid =␣
↪False)
```

[18]:
```
<Axes: title={'center': 'attend'}, xlabel='month'>
```

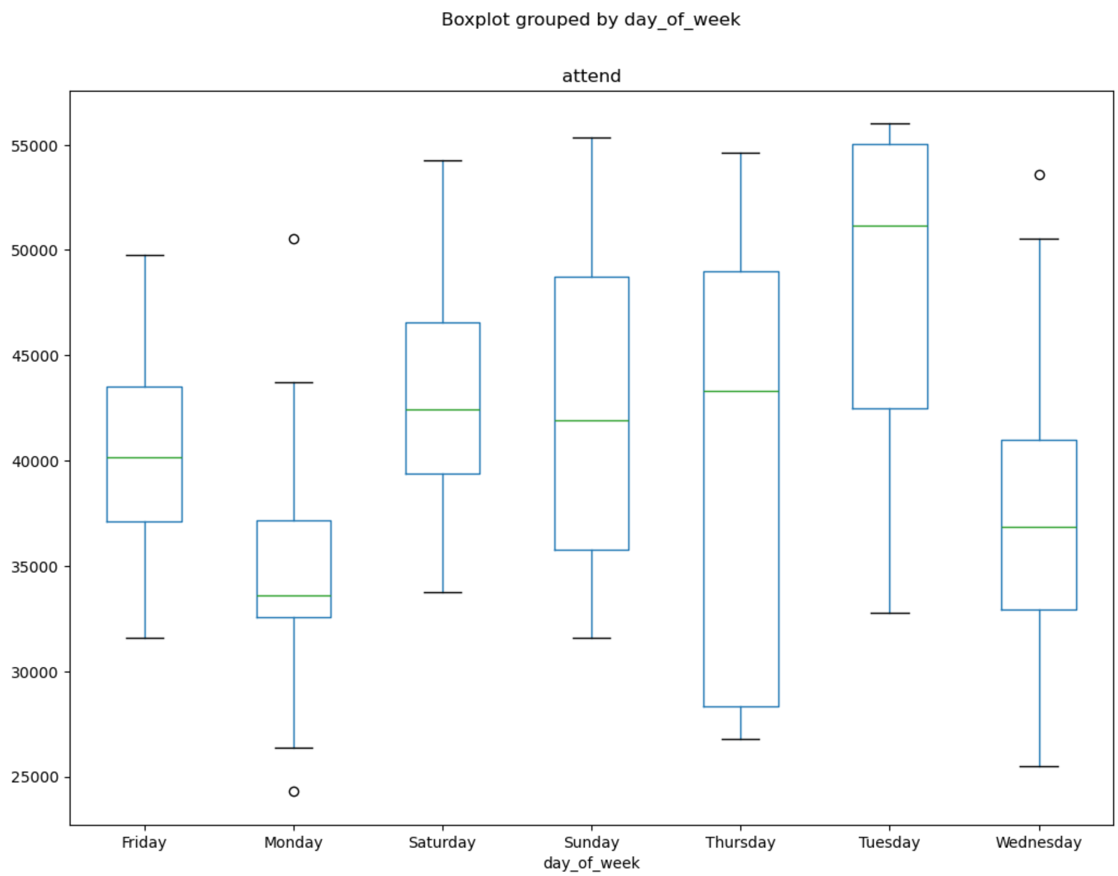



The above boxplot illustrates the highest monthly median attendance being in June, with a consistently higher attendance than the rest of the months. October on the other hand has the lowest median and the most consistently lower attendance. April and July have a similar median and wide range of attendance, but overall July attendance is greater.

[28]:
```
# Boxplot of daily attendance
```

```
dodgers_data.boxplot(by ='day_of_week', column =['attend'], figsize=(12,9),␣
↪grid = False)
```

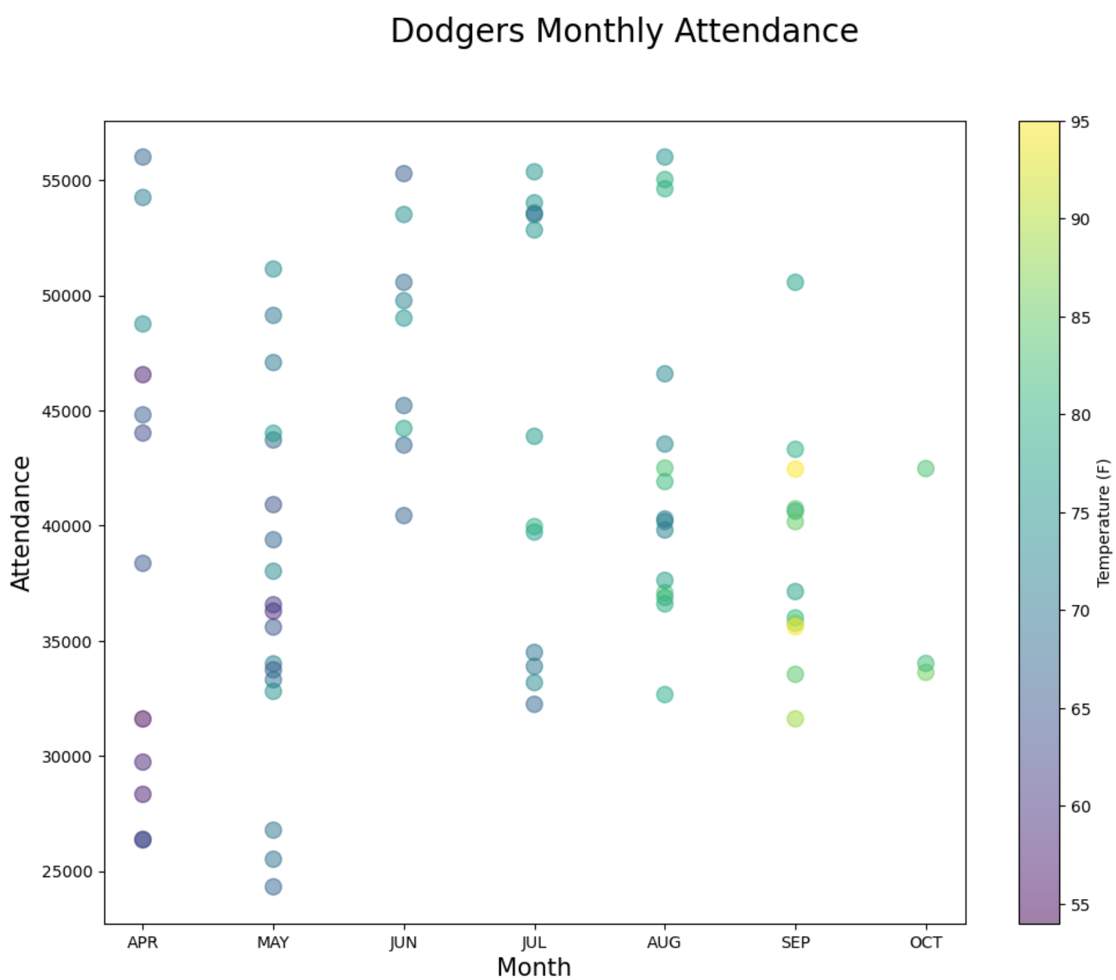
[28]: <Axes: title={'center': 'attend'}, xlabel='day_of_week'>




This boxplot shows us that the highest daily median attendance is on Tuesdays, with a consistently higher attendance than the rest of the days. Mondays have the lowest median and consistently lower attendance. At this point, now that I have an idea about monthly and daily averages, it will help to visualize what other factors could have an effect on attendance.

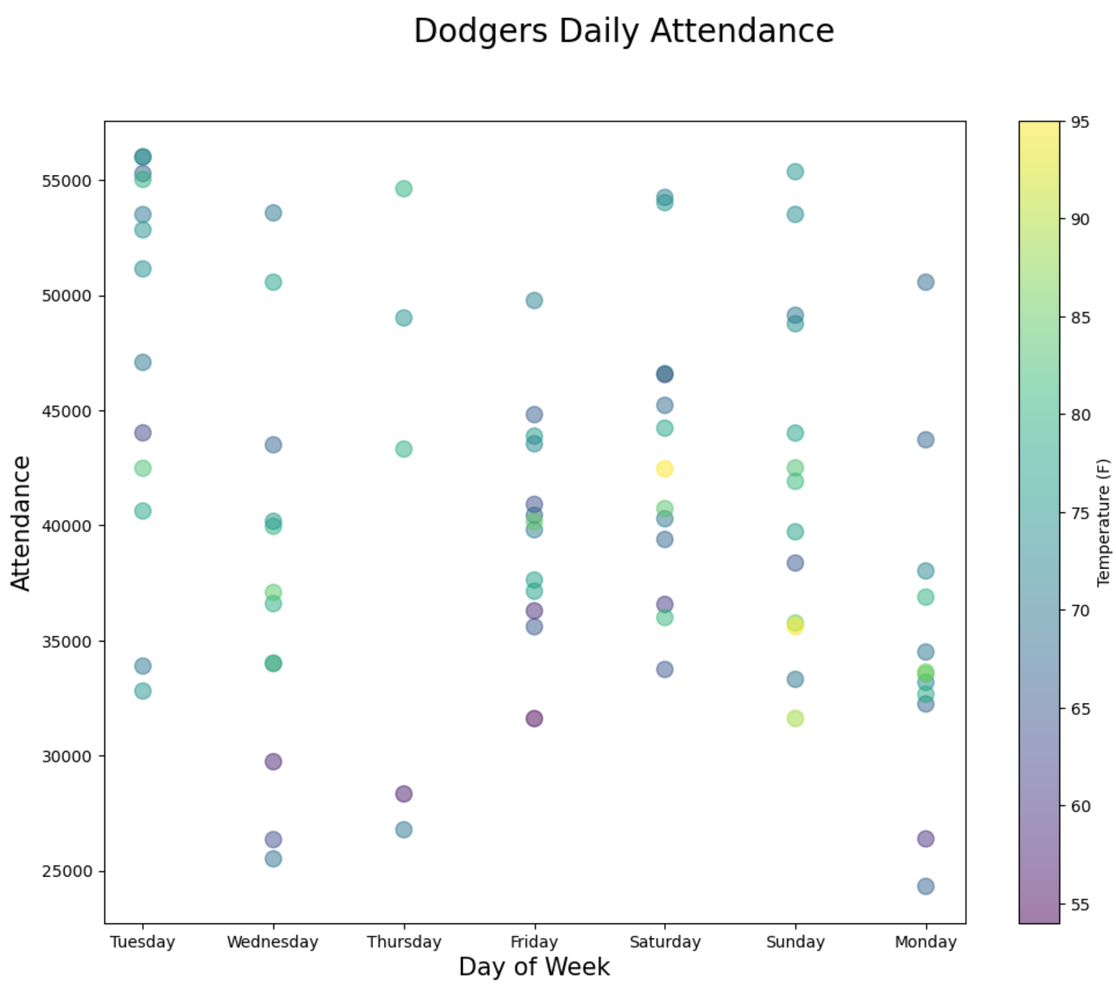
[31]:
```
# Scatterplot of monthly attendance and temperature

plt.figure(figsize=(12, 9))
plt.scatter('month', 'attend', s=100, c='temp', alpha=0.5, data=dodgers_data)
plt.xlabel('Month', fontsize=15)
plt.ylabel('Attendance', fontsize=15)
plt.suptitle('Dodgers Monthly Attendance', fontsize=20)
plt.colorbar(label='Temperature (F)')
plt.show()
```




[33]:
```
# Scatterplot of daily attendance and temperature

plt.figure(figsize=(12, 9))
plt.scatter('day_of_week', 'attend', s=100, c='temp', alpha=0.5,
    data=dodgers_data)
plt.xlabel('Day of Week', fontsize=15)
plt.ylabel('Attendance', fontsize=15)
plt.suptitle('Dodgers Daily Attendance', fontsize=20)
plt.colorbar(label='Temperature (F)')
plt.show()
```

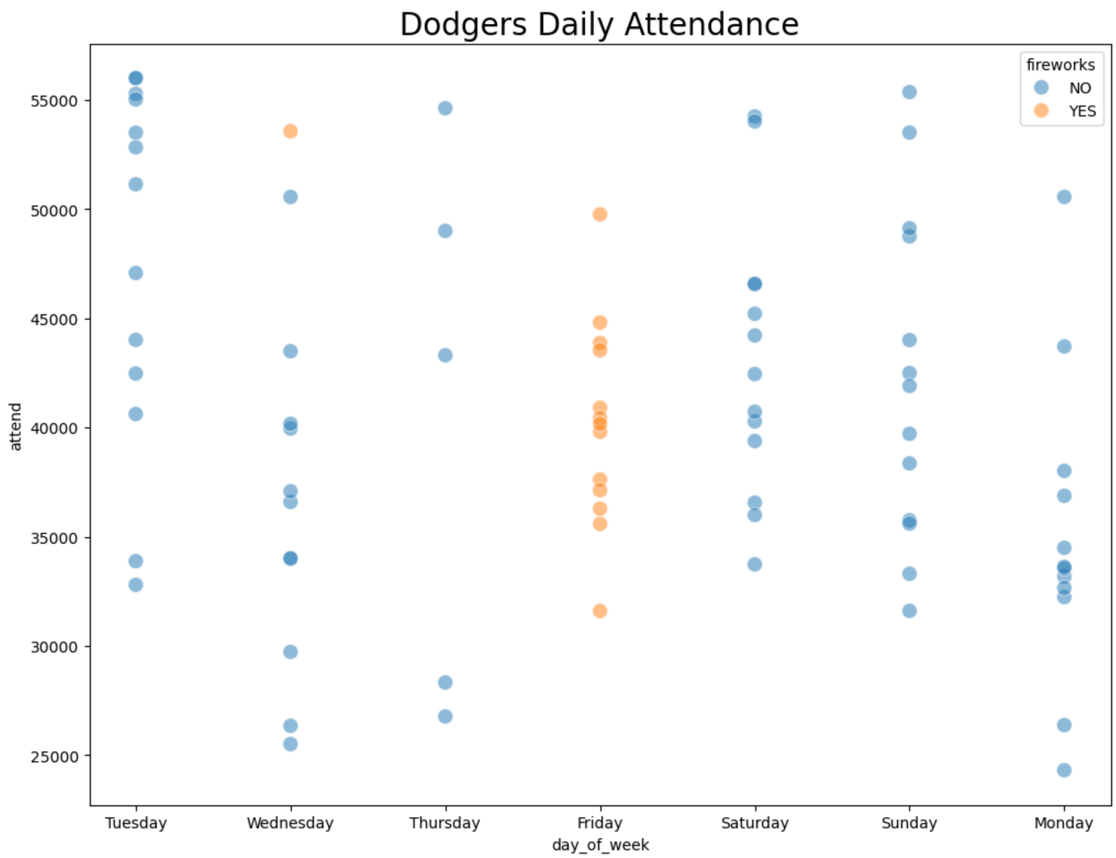



[35]:
```
# Scatterplot of daily attendance and fireworks

f, ax = plt.subplots(figsize=(12, 9))

g = sns.scatterplot(x='day_of_week', y='attend', s=100, alpha=0.5,
              hue='fireworks',
              data=dodgers_data);
plt.title('Dodgers Daily Attendance', fontsize=20)
g.set()
```

[35]: []




Looking at the scatterplots of attendance by week day and month, with a few additional factors taken into consideration using colors, it looks like timing is most important. Neither temperature nor fireworks appear to have a large effect on attendance.<./h4>

Recommendations to Management:

Based on the analysis, management can implement strategies such as:

Targeted Promotions: If certain promotional items consistently drive higher attendance (e.g., bobbleheads), prioritize these for key games or less popular days.

Weekend Maximization: Capitalize on higher weekend attendance by scheduling marquee matchups or special events on Saturdays and Sundays.

Weekday Incentives: Introduce attractive promotions or lower ticket prices on weekdays, especially Mondays and Tuesdays, which often show lower attendance.

Rivalry Games: Strategically schedule games against popular or rival teams, as these often naturally draw larger crowds. Seasonal Planning: Identify months with historically lower attendance and consider introducing unique events or promotions during those periods to boost interest.

[ ]: